



Dinul Iman

Full-stack Developer · Jakarta, Indonesia

Full-stack developer focused on the backend, with a love for working with data — from payroll systems to ETL pipelines.

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ABOUT

Hi! I'm Dinul Iman, an Information Technology graduate from Universitas Sumatera Utara focused on full-stack development, with my strongest experience in backend and databases. Day to day I work with PHP, TypeScript, Node.js and relational databases such as PostgreSQL and Oracle.

My most notable work is the Payroll Management System I worked on for PT PELNI as a programmer at PT Solusi Integrasi Utama, a PELNI vendor. It serves around 4,800–5,000 employees — from PPh 21 income tax, BPJS, pension and cooperative deductions to journal integration with Oracle. A favourite achievement: cutting the payroll run from more than 10 minutes to under 1 minute by optimizing stored procedures.

I enjoy work that sits close to data: ETL/ELT, data warehousing and automation. Lately I've also been actively exploring AI coding agents such as Claude Code, Codex and MCP.

TARGET ROLES

Backend Developer / Engineer

Full-stack Developer

Software Engineer

Data Engineer

HRIS / Payroll Technology

EXPERIENCE

Programmer — PT Solusi Integrasi Utama (PT PELNI vendor)

Aug 2025 — Present

- PT PELNI Payroll Management System ($\pm 4,800$ –5,000 employees): development with PHP/CodeIgniter and PostgreSQL.
- Payroll calculation and processing: PPh 21 income tax, BPJS, pension, cooperative, receivables and salary proration.
- Worked on the data integration Oracle $\rightarrow$ PostgreSQL $\rightarrow$ payroll $\rightarrow$ Oracle journal.
- Optimized payroll stored procedures, cutting a run that took >10 minutes to <1 minute.
- Built payslips and payroll reports with JasperReports/JasperServer.
- PT PELNI ticketing application: backend development with NestJS and PostgreSQL, including the shipping insurance module.
- Used Docker / Rancher Desktop for application environments.

Data Structures & Algorithms Lab Assistant — Universitas Sumatera Utara

Jan 2021 — Jun 2021

- Teaching assistant for the Data Structures & Algorithms lab course.

Database Lab Assistant — Universitas Sumatera Utara

Jun 2020 — Jan 2021

- Teaching assistant for the Database lab course.

EDUCATION

B.Sc. Information Technology — Universitas Sumatera Utara (USU)

GPA 3.50 · Lab assistant for Databases and for Data Structures & Algorithms.

SKILLS

PHP & CodeIgniter	★★★★★	Laravel	★★★★☆
Yii2	★★★☆☆	Node.js & NestJS	★★★★☆
TypeScript	★★★★☆	Python & Django	★★★☆☆
Go & Gin	★★★☆☆	JavaScript	★★★★☆
React & Next.js	★★★☆☆	PostgreSQL	★★★★★
Oracle	★★★★☆	MySQL	★★★★☆
MongoDB	★★★☆☆	ETL & Data Integration	★★★★☆
JasperReports / JasperServer	★★★★☆	Tableau	★★★☆☆
BigQuery	★★★☆☆	Airflow & dbt	★★★☆☆
Spark & Kafka	★★★☆☆	Docker / Rancher	★★★☆☆
Payroll & HRIS	★★★★★		

PROJECTS

Payroll Management System for PT PELNI

Payroll system for ±4,800–5,000 employees: PPh 21 income tax, BPJS, pension, cooperative, receivables and proration; Oracle ↔ PostgreSQL integration; payslips & reports via JasperServer. Stored procedure optimization cut the run from >10 minutes to <1 minute. Built as a programmer at PT Solusi Integrasi Utama.

PHP

CodeIgniter

PostgreSQL

Oracle

JasperServer

Docker

Ticketing Application for PT PELNI

Backend for a ticketing application, including a shipping insurance module. Built at PT Solusi Integrasi Utama.

NestJS

TypeScript

PostgreSQL

Data Engineering Portfolio

Three end-to-end data pipelines built like small production systems, with CI: batch ELT of weather data for 10 Indonesian cities (Airflow, MinIO, PostgreSQL, dbt, Metabase), real-time e-commerce order streaming (Kafka, Spark Structured Streaming with watermarks, dead-letter queue, Grafana), and an NYC Taxi bronze → silver → gold lakehouse (PySpark, Delta Lake, quality gates, time travel).

[Python](#)[Airflow](#)[dbt](#)[Spark](#)[Kafka](#)[Delta Lake](#)[PostgreSQL](#)[Docker](#)

Naver SmartStore Scraper API

[link](#)

REST API that extracts product data from Naver SmartStore pages using a stealth headless browser, rotating proxies & fingerprints, throttling and caching. The scraping backend can be swapped via an env var with no code changes.

[TypeScript](#)[Node.js](#)[Playwright](#)

GoBack — E-Commerce Platform

Full-stack e-commerce app: Go backend with JWT auth, product & stock management, addresses, Midtrans payments and an admin dashboard; React frontend.

[Go](#)[Gin](#)[GORM](#)[PostgreSQL](#)[React](#)[JWT](#)

Job Market Analytics

Job market analysis platform: job-posting scraping, automatic skill extraction, in-demand skills, salary statistics by location & level, and career path recommendations.

[Python](#)[Flask](#)[Vue.js](#)

Botto — RAG Chatbot

Chatbot for asking questions about your own documents (PDF, Word, TXT) using Retrieval-Augmented Generation with streaming responses. The whole stack is free.

[Next.js](#)[TypeScript](#)[LangChain.js](#)[ChromaDB](#)[Groq \(Llama 3.1\)](#)

Toko Syuhada — Bicycle Shop Website

Bicycle catalog & sales website with category filters, product search, detailed specs and articles.

[Next.js](#)[TypeScript](#)[Prisma](#)[PostgreSQL](#)[Tailwind CSS](#)

Machine Learning Experiments

[link](#)

A collection of notebooks: text sentiment analysis, ETH price forecasting with LSTM, a seq2seq chatbot, real-time apple/tomato image classification via webcam, and data analysis (car prices, ad campaigns, diabetes).

[Python](#)[TensorFlow/Keras](#)[Pandas](#)[OpenCV](#)

CV Quest

[link](#)

An interactive CV as a browser RPG — yes, the game you're playing right now!

[JavaScript](#)[Canvas](#)[HTML/CSS](#)

INTERESTS

AI & Data · ETL/ELT & Data Warehousing · TypeScript · AI Coding Agents · Claude Code · Codex · MCP · Automation